

EXPERIMENT 5

Aim

To implement **Support Vector Machine (SVM)** for classification on the Titanic dataset and improve model performance using **Hyperparameter Tuning (GridSearchCV)**.

Theory:

Support Vector Machine (SVM) is a supervised machine learning algorithm used for classification and regression problems.

In classification, SVM works by finding an **optimal hyperplane** that separates data points of different classes with the maximum margin.

The main idea of SVM is:

- Find the best boundary that separates classes.
- Maximize the distance (margin) between the nearest data points of both classes.
- These nearest points are called **Support Vectors**.

For non-linear data, SVM uses a **Kernel Trick** such as:

- Linear
- RBF (Radial Basis Function)
- Polynomial

Important Hyperparameters:

1. C (Regularization Parameter)

Controls trade-off between margin size and classification error.

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2. Kernel

Defines type of decision boundary.

3. Gamma

Controls how far the influence of a single training example reaches.

Hyperparameter tuning helps in selecting the best combination of these parameters to improve model performance.

Real-Life Example

Imagine you are separating:

- Apples

- Oranges

Based on:

- Weight
- Color
- Size

Now imagine drawing a line on a table that separates apples on one side and

oranges on the other. SVM tries to:

- Draw the best possible line
- Keep maximum distance from both fruits
- Make sure fruits are correctly separated

If fruits are mixed in a curved pattern, SVM uses a curved boundary (like RBF kernel). Hyperparameter tuning is like: Trying different types of lines and settings to find the best separation.

Choosing Dataset:

# PassengerId	# Survived	# Pclass	# Name	# Sex	# Age	# SibSp
Passenger ID	Whether Survived or not: 0 = No, 1 = Yes	Ticket class: 1 = 1st, 2 = 2nd, 3 = 3rd	Name of the Passenger	Gender	Age in Years	No. of siblings aboard the Tita
			891 unique values	male female 35%		
1	0	3	Braund, Mr. Owen Harris	male	22	1
2	1	1	Coombs, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1
3	1	3	Hekkinen, Miss. Laina	female	26	0
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1
5	0	3	Allen, Mr. William Henry	male	35	0

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We choose the titanic Dataset for this experiment.

The sinking of the Titanic is one of the most infamous shipwrecks in history.

While there was some element of luck involved in surviving, it seems some groups of people were more likely to survive than others.

In this challenge, we build a predictive model that answers the question: “what sorts of people were more likely to survive?” using passenger data (ie name, age, gender, socio-economic class, etc).

Step 1: Data Collection

Download Titanic dataset from Kaggle.
Use train.csv.

Step 2: Data Preprocessing

Handle Missing Values:

Age filled using median.

Embarked filled using mode.

Remove Unnecessary Columns:

PassengerId

Name

Ticket

Cabin

Encode Categorical Data:

Sex converted to numeric (male=0, female=1)

Embarked converted using one-hot encoding

Separate:

Features (X)

Target (Survived)

Step 3: Train-Test Split

80% for training

20% for testing

Ensures unbiased evaluation

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Step 4: Feature Scaling

StandardScaler used

Important because SVM is sensitive to feature magnitudes

Step 5: Model Training (Baseline SVM)

Train SVM using default parameters
Predict test data
Calculate baseline accuracy

Step 6: Hyperparameter Tuning

Use GridSearchCV
Try different combinations of:
C
Kernel
Gamma
5-fold cross-validation used
Select best parameter combination

Step 7: Model Evaluation

Evaluate using:
Accuracy Score
Confusion Matrix

Precision
Recall
F1-Score

CODE :

```
# =====  
  
# SVM Classification with Hyperparameter Tuning # Dataset: Titanic (Kaggle)  
# ===== # Step 1:  
Import Libraries import numpy as np  
import pandas as pd  
  
import matplotlib.pyplot as plt import seaborn as sns  
  
from sklearn.model_selection import train_test_split, GridSearchCV from  
sklearn.preprocessing import StandardScaler  
from sklearn.svm import SVC  
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```

```
from sklearn.metrics import accuracy_score, confusion_matrix,  
classification_report
```

```
# Step 2: Load Dataset (Correct Path)
```

```
data = pd.read_csv('sample_data/train.csv')
```

```
# Step 3: Basic Data Cleaning
```

```
# Fill missing Age with median
```

```
data['Age'] = data['Age'].fillna(data['Age'].median())
```

```
# Fill missing Embarked with mode
```

```
data['Embarked'] = data['Embarked'].fillna(data['Embarked'].mode()[0])
```

```
# Drop unnecessary columns
```

```
data = data.drop(['PassengerId', 'Name', 'Ticket', 'Cabin'], axis=1)
```

```
# Convert categorical variables
```

```
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
```

```
# One-hot encoding for Embarked
```

```
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
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```

```
# Step 4: Define Features and Target
```

```
X = data.drop('Survived', axis=1) y = data['Survived']
```

```
# Step 5: Train-Test Split
```

```
X_train, X_test, y_train, y_test = train_test_split( X, y, test_size=0.2,
random_state=42
)
```

```
# Step 6: Feature Scaling (Very Important for SVM) scaler = StandardScaler()
X_train = scaler.fit_transform(X_train) X_test = scaler.transform(X_test)
```

```
# Step 7: Baseline SVM Model svm_model = SVC() svm_model.fit(X_train,
y_train)
```

```
y_pred = svm_model.predict(X_test)
```

```
print("Baseline Accuracy:", accuracy_score(y_test, y_pred))
```

```
# Step 8: Hyperparameter Tuning using GridSearchCV param_grid = {
'C': [0.1, 1, 10, 100],
```

```
'gamma': ['scale', 'auto'],
```

```
'kernel': ['linear', 'rbf']
```

```
}
```

```
grid = GridSearchCV(SVC(), param_grid, cv=5, verbose=1) grid.fit(X_train,
```

```
y_train)
```

```
print("\nBest Parameters Found:", grid.best_params_)
```

```
# Step 9: Evaluate Tuned Model best_model = grid.best_estimator_  
Prakh y_pred_best =
```

```
best_model.predict(X_test)
```

```
print("\nTuned Model Accuracy:", accuracy_score(y_test, y_pred_best))
```

```
print("\nConfusion Matrix:\n", confusion_matrix(y_test, y_pred_best))
```

```
print("\nClassification Report:\n", classification_report(y_test, y_pred_best))
```

OUTPUT :

```
Baseline Accuracy: 0.8212290502793296  
Fitting 5 folds for each of 16 candidates, totalling 80 fits  
  
Best Parameters Found: {'C': 1, 'gamma': 'scale', 'kernel': 'rbf'}  
  
Tuned Model Accuracy: 0.8212290502793296  
  
Confusion Matrix:  
[[95 10]  
 [22 52]]  
  
Classification Report:  
              precision    recall  f1-score   support  
  
    0           0.81       0.90       0.86         105  
    1           0.84       0.70       0.76          74  
  
   accuracy              0.82         179  
  macro avg           0.83       0.80       0.81         179  
 weighted avg           0.82       0.82       0.82         179
```

Results Summary:

Baseline Accuracy: 82.12%

Tuned Accuracy: 82.12%

Best Parameters:

$C = 1$

Kernel = RBF

Gamma = scale

The tuning confirmed that default parameters were already optimal.

CONCLUSION:

The Support Vector Machine (SVM) classifier was successfully implemented on the Titanic dataset. Data preprocessing steps such as handling missing values, encoding categorical variables, and feature scaling were performed to prepare the dataset for modeling.